

Deep Learning for Vessel Segmentation in Cerebral CTA

Midterm Project Proposal

Abstract

Accurate segmentation of cerebral vasculature from computed tomography angiography (CTA) is essential for the diagnosis and surgical planning of stroke, intracranial aneurysms, and other cerebrovascular diseases. Despite significant progress in deep learning for medical image segmentation, standard overlap-based training objectives such as Dice and cross-entropy can produce segmentations that score well on aggregate metrics yet contain topological failures, including severed vessel connectivity, that render them clinically unusable. Topology-preserving loss functions such as centerline Dice (clDice) and Skeleton Recall have been proposed to address this, but existing evaluations either introduce them alongside architectural changes or report only aggregate scores from multi-team challenge leaderboards, making it difficult to isolate the effect of the loss function itself.

This project presents a controlled ablation study isolating the impact of topology-aware loss functions on cerebral CTA vessel segmentation. We train a 3D U-Net from scratch in PyTorch on the TopCoW 2024 dataset (125 expert-annotated CTA volumes with voxel-level labels for 13 Circle of Willis components) under three loss configurations: (1) Dice + cross-entropy as a baseline, (2) Dice + cross-entropy + soft-clDice, and (3) Dice + cross-entropy + Skeleton Recall. Architecture, preprocessing, patch sampling, augmentation, and all hyperparameters are held identical across runs; we select the 3D U-Net specifically because it remains the strongest baseline under controlled comparison, allowing any performance difference to be attributed solely to the loss formulation rather than confounded by architectural variables.

Beyond the standard multi-metric evaluation protocol (DSC, clDice, HD95, Betti-0 errors), we contribute two additional analyses absent from the existing literature. First, we provide a stratified per-vessel-class breakdown using the 13 anatomical labels from TopCoW, grouping results into large named arteries (e.g., ICA, MCA, BA) and small communicating segments (Acom, Pcom). We hypothesize that topology-aware losses disproportionately improve segmentation of the thin communicating arteries that are most clinically significant for aneurysm and collateral flow assessment yet most prone to topological failure. Second, we report computational cost profiling (wall-clock training time, peak GPU memory, time per epoch) across all three configurations, providing the first direct resource comparison of clDice and Skeleton Recall on volumetric CTA data. Finally, drawing on the author's clinical experience as a neurosurgical navigation specialist, we include qualitative 3D renderings of select validation cases to illustrate segmentation failures that are invisible to aggregate metrics but consequential for surgical planning.

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